SMI Nudge: A Model-Agnostic Post-Processing Framework for Retail Demand Forecasting

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***Abstract*—Retail demand forecasting is necessary for effective inventory management, yet traditional models fail to keep up with rapid market shifts. To address this, the SMI Nudger is a model-agnostic rule-based post-processing block that enhances baseline predictions from statistical models (ARIMA, STL-ARIMA, and Prophet) and deep learning (LSTM and Chronos) using Sales Momentum Index (SMI) signals without retraining. SMI derives three signals directly from sales history — Hype Index (spike intensity), Momentum (trend acceleration), and Persistence (trend stability) — to capture recent demand trends.**

The system demonstrates significant MASE and WQL improvements on structured Walmart data (Chronos with SMI: 0.0449 absolute MASE reduction), with marginal but consistent gains on sparse M5 series where STL-ARIMA remains competitive. Results confirm that SMI nudging improves forecast accuracy while preserving model ranking. This work contributes a transparent, practical enhancement framework that bridges the gap between static predictions and dynamic demand patterns.

***Index Terms*—Retail Demand Forecasting, Time-Series Analysis, Sales Momentum Index**

I. Introduction

Organizations and companies need accurate forecasting systems that adapt fast and are operationally interpretable. But existing forecasts do not quickly reflect sudden demand changes, and many of the existing models are opaque, which offer no interpretable results for business users. Also, frequent retraining is computationally expensive and unsustainable. Current work exposes a clear gap that the framework bridges directly. No existing system derives attention signals from recent historical retail sales data and applies them through a model-agnostic post-processing layer without retraining. This gap is addressed through three contributions:

A model-agnostic, retraining-free Nudger that improves both statistical and deep learning models' forecasts using a single unified rule set.

A Sales Momentum Index (SMI) signals — Hype Index [1] [2] [3] [4], Momentum, and Persistence — three interpretable signals derived purely from recent historical retail sales data, newly combined for retail demand correction.

A horizon-sensitive, rule-based external block that uses decaying multiplicative factors and bounded safety multipliers (3), validated on both structured and sparse retail benchmarks.

A. Organisation of the Paper

The remainder of this paper is organized as follows: Section II reviews related work in retail forecasting and momentum-based signal research. Section III details the system architecture, datasets, models, SMI signal design, and the Nudger framework. Section IV presents experimental results and analysis across both datasets. Section V concludes the paper and outlines future directions [5].

II. Literature Review

Retail demand forecasting has been extensively studied across statistical, machine learning, and hybrid paradigms. Classical time series models such as ARIMA (Auto-Regressive Integrated Moving Average) [6] and its seasonal-trend decomposition variant STL-ARIMA (Seasonal Trend decomposition using Loess-AutoRegressive Integrated. Moving Average) [7] remain competitive baselines due to their interpretability and low computational cost, as confirmed by Kachmar et al [8] who benchmarked ARIMA and LSTM (Long Short-Term Memory) on retail sales. Fildes et al [9] provide a comprehensive survey of retail forecasting practice, highlighting the persistent gap between model accuracy and operational adaptability — the central problem this framework addresses. Petropoulos et al [10] further contextualise these methods within the broader forecasting landscape, establishing the theoretical grounding for model selection in this work. Deep learning approaches have substantially advanced probabilistic retail forecasting. Hochreiter and Schmidhuber's [11] LSTM laid the foundation for recurrent sequence modelling, while Hewamalage et al [12] systematically reviewed RNN-based forecasters, identifying robustness limitations on sparse retail series. Salinas et al [13] introduced DeepAR, establishing autoregressive recurrent networks as a canonical reference for probabilistic forecasting with quantile evaluation — the WQL (Weighted Quantile Loss) metric used here. Lim et al [14] extended this line with Temporal Fusion Transformers, demonstrating interpretable multi-horizon forecasting that motivates the horizon-sensitive Nudger design. Falatouri et al [15] directly compare SARIMA (Seasonal AutoRegressive Moving Average) and LSTM in retail supply chains, reinforcing the motivation for hybrid correction over single-model reliance. Pretrained foundation models for time series have recently emerged as strong zero-shot forecasters. Ansari et al [16] introduced Chronos, a language-model-based probabilistic forecaster evaluated under the M5 benchmark; Arango et al [17] further extended it with exogenous variable support. The M5 competition results [18] and Spiliotis et al [19] provide the reference performance landscape for both structured and sparse SKU (Stock Keeping Units) - level series, demonstrating that intermittent demand remains an open challenge even for state-of-the-art models. Taylor and Letham's Prophet [20] completes the baseline ensemble as a scalable additive decomposition model. The SMI signals derive inspiration from market attention and social signal research. Cao et al [1] formalise the Hype Index as an NLP-driven measure of market attention, which is adapted into a sales-domain spike intensity metric. Badulescu et al [2] and Kazemian et al [3] investigate social media signals for demand modelling and lead-lag forecasting, respectively, validating momentum-based correction as a viable forecasting supplement. Putri et al [4] demonstrate exponential smoothing in supply chain bullwhip reduction, directly informing the Persistence signal design. Huber and Stuckenschmidt [21] show that calendric and promotional events drive structured demand spikes in retail — the primary target of the Nudger's bullish rules [5].

III. Methodology

A. System Architecture

The system is a continuous pipeline across five stages. Raw sales data from the retail sales dataset is parsed, duplicate sales are removed, it is resampled to a fixed frequency, and missing values are filled using linear interpolation. The preprocessed series is then passed independently to each baseline model: ARIMA, STL-ARIMA, LSTM, Prophet, and Chronos. Each model produces a 12-week-ahead probabilistic forecast without any modification to its internal parameters. Parallelly, the SMI computation block reads the last 8 weeks of training-window sales history and derives three signals — Hype Index [1] [2] [3] [4], Momentum, and Persistence — strictly from observed data to prevent leakage. The Nudger receives both the baseline forecast and the three SMI signals. It identifies the appropriate rule depending on which week's forecast it is nudging and if the SMI signals were in bearish condition or bullish condition, computes a confidence-weighted (1), decay-adjusted multiplier (3) for each forecast week, and applies bounded multiplicative corrections to produce the nudged output. The evaluation block compares baseline and nudged forecasts against held-out actuals using MASE (Mean Absolute Scaled Error) [17] [8] and WQL [17], alongside computational metrics (memory, compilation time, FLOPs (Floating Point Operations)) reported independently of nudging.

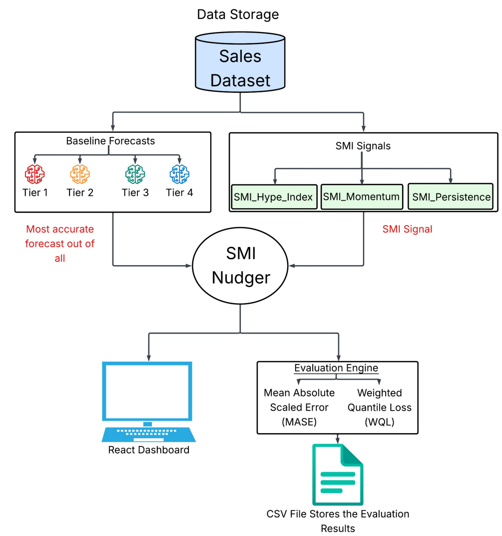


Fig. 1: System Architecture: Baseline $\rightarrow$ SMI $\rightarrow$ Nudger $\rightarrow$ Evaluation Pipeline

B. Datasets

The SMI–Nudger was tested on two benchmark datasets across contrasting demand conditions. The Walmart Recruiting Store Sales Forecasting Dataset [1] covers 45 stores across 99 departments at weekly frequency (February 2010 – October 2012), with rich covariates including CPI, unemployment, fuel price, and five promotional markdown flags. Its structured periodicity and dense coverage make it ideal for evaluating momentum-driven demand surges from known promotional events. The M5 Forecasting Accuracy Dataset [2] contains daily item-level sales for 3,049 SKUs across ten stores over five years (January 2011 – June 2016), characterised by high sparsity, frequent zero-sales days, and intermittent demand. The M5 Forecasting Accuracy Dataset was aggregated to weekly frequency by summation before modelling to preserve weekly seasonality while reducing daily noise.

C. Models Used

An ARIMA model with order (7, 0, 1) was fit directly on the raw weekly sales series using the statsmodels library. The model receives a univariate time series as input and produces a 12-step-ahead point forecast.

[1] https://www.kaggle.com/competitions/walmart-recruiting-store-sales-forecasting/overview

[2] https://www.kaggle.com/competitions/m5-forecasting-accuracy/data

STL was applied to the training series over 52 weeks. An ARIMA(1,1,1) model was then fitted to the deseasonalized signal, comprising the trend and residual components; forecasts were generated 12 steps ahead, and the seasonal component was reintroduced by tiling the final 52 seasonal values, thereby reconstructing the full forecast on the original scale.

Meta's Prophet model was applied with annual seasonality enabled and weekly and daily seasonality disabled, consistent with the weekly granularity of the input data. The model receives a dataframe with columns date, anchored to weekly Friday frequency, and sales, and extracts the forecasted sales column from a 12-week future dataframe as the point forecast.

A single-layer LSTM network was implemented from scratch in PyTorch, comprising an LSTM layer with hidden size 64 followed by a linear projection to a scalar output. The model was trained on sliding window sequences of length 12 drawn from a StandardScaler-normalised univariate series, optimised for 200 epochs using Adam and MSE Loss (Mean Squared Error Loss); inference was performed autoregressively before inverse-transforming to the original scale.

Chronos was loaded using the chronos-t5-small checkpoint. The input series was scaled using a MinMax scheme with a log-scale heuristic applied to high-range series such as Walmart sales data; probabilistic inference was performed with 20 Monte Carlo samples at a temperature of 0.7 under, and the mean across samples was taken as the point forecast before inverse-transforming to the original scale.

D. Post-Processing Framework

Baseline forecasts are adjusted through a lightweight post-processing layer, the Nudger, which applies a bounded multiplicative correction without modifying the underlying model. The Nudger uses SMI signals, three signals derived from recent historical retail sales data alone.

Hype Index: How intense is the current demand spike relative to the historical baseline?

Momentum: Is demand accelerating or decelerating right now?

Persistence: Is this a fleeting spike, or a sustained trend? At each forecast origin, the Nudger reads the last 8 weeks of SMI signals, determines the active demand rule (bullish, bearish, or neutral), and applies a small bounded multiplier (3) to the baseline forecast. Fig. 1 shows the end-to-end pipeline from baseline forecast through SMI signals usage, rule-based nudging, and nudged forecast output.

E. SMI Signals

Three momentum-inspired signals are engineered from recent historical retail sales data, computed strictly on training data to prevent leakage.

The Hype Index captures normalised spike intensity. For each week where observed sales exceed the centred 4-week rolling mean, a scale-aware magnitude is computed as that week's sales divided by the global weekly mean. This magnitude is propagated with asymmetric weights — 0.1, 0.5, 0.9, and 0.5 — across the two-week pre-spike window, one-week pre-spike window, spike week, and immediate post-spike week respectively. The aggregated signal is min-max normalised to [0, 1].

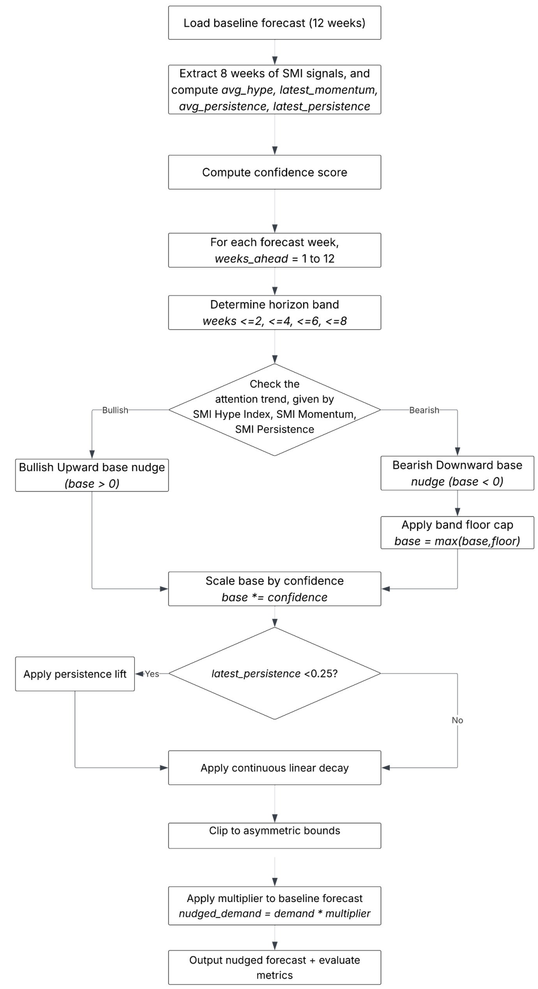


Fig. 2: End-to-end integration: Baseline Forecast → SMI signal extraction → Rule-based nudging → Nudged forecast

Momentum is the first-order difference of the Hype Index over consecutive weeks. A positive value signals growing demand intensity; a negative value signals fading enthusiasm.

Persistence is the Exponential Moving Average (EMA) of the Hype Index with a span of four weeks. It distinguishes transient spikes (high hype, low persistence) from sustained elevation (high hype, high persistence), which is critical for calibrating nudge magnitude.

Algorithm 1 describes the generation of SMI-based signals.

F. SMI Signals Graphs

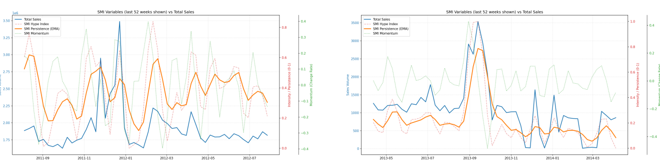
Fig. 3a shows SMI signals over the last 52 weeks of the Walmart dataset. The Hype Index closely tracks the prominent demand surge, while Momentum oscillates between positive and negative, reflecting repeated acceleration shifts across the period.

Fig. 3b shows SMI signals over the last 52 weeks of the M5 dataset. The Hype Index captures a sharp surge around

Algorithm 1 Sales Momentum Index (SMI) Signal Generation

Require: Weekly sales time series $total\_sales$, weights $w = [0.1, 0.5, 0.9, 0.5]$, EMA span $= 4$
Ensure: SMI_{Hype}, $SMI_{Momentum}$, $SMI_{Persistence}$

$df_{weekly} \leftarrow$ resample($total\_sales$, weekly, Friday-ending)
$rolling\_mean \leftarrow$ CenteredRollingMean(df_{weekly}, window=4)
$global\_mean \leftarrow$ Mean(df_{weekly})
Initialize $smi\_signal \leftarrow \mathbf{0}$
for $i = 0$ to $|df_{weekly}| - 1$ **do**
  if $df_{weekly}[i] > rolling\_mean[i]$ **then**
    $mag \leftarrow df_{weekly}[i]/global\_mean$
    $smi\_signal[i-2] \mathrel{+}= w_1 \cdot mag$
    $smi\_signal[i-1] \mathrel{+}= w_2 \cdot mag$
    $smi\_signal[i] \mathrel{+}= w_3 \cdot mag$
    $smi\_signal[i+1] \mathrel{+}= w_4 \cdot mag$
  end if
end for
$SMI_{Hype} \leftarrow$ MinMaxNormalize($smi\_signal + \mathcal{U}(0, \epsilon)$)
$SMI_{Momentum} \leftarrow$ Difference(SMI_{Hype})
$SMI_{Persistence} \leftarrow$ EMA(SMI_{Hype}, span=4)
return SMI_{Hype}, $SMI_{Momentum}$, $SMI_{Persistence}$



(a) Walmart Dataset　　(b) M5 Dataset

Fig. 3: SMI Signals across the two Datasets

early September 2013 with high fidelity. Persistence reflects a lag-based carry-over effect. Momentum indicating demand deceleration even when absolute hype remains elevated.

G. Nudger: Confidence, Horizon Sensitivity, and Decay

At each forecast origin, the Nudger extracts hype volatility, momentum strength, and persistence score from the last 8 weeks of SMI signals and computes a confidence score (1):

$$confidence = 0.5 + 0.5 \times \min\left(1.0,\ \frac{persistence\_score}{0.22} + \frac{momentum\_strength}{0.12} + \frac{hype\_volatility}{0.10}\right) \quad (1)$$

where $hype\_volatility$ is the standard deviation of the SMI Hype Index over the last 8 weeks

$momentum\_strength$ is the absolute latest momentum, and

$persistence\_score$ is the average persistence

This yields a score in $[0.5, 1.0]$, where higher values reflect stronger, more consistent momentum signals. The intermediate scaled base is calculated as:

$$base = base \times confidence \quad (2)$$

where $base$ is the base nudge value taken from the horizon-dependent regime, and

$confidence$ is the confidence multiplier computed from SMI signals.

The final horizon-specific multiplier (3) is:

$$multiplier = 1.0 + (base \times \delta(h)) \quad (3)$$

where $multiplier$ is the final adjustment multiplier applied to the baseline forecast, hard-clamped to $[0.978, 1.095]$, and

$\delta(h)$ is the horizon-aware decay factor defined in (4) which is computed as:,

$$\delta(h) = \max(0,\ 1 - (h - 0.5) \times 0.11) \quad (4)$$

where h is the forecast horizon in weeks,

$confidence$ (1) is the confidence score computed from SMI signals, and

$multiplier$ (3)is the final adjustment multiplier applied to the baseline forecast

The multiplier (3) is hard-clamped to $[0.978, 1.095]$. Table I summarises the demand rules, horizon-dependent thresholds, nudge magnitudes, and decay factors applied at each forecast band.

TABLE I: Nudger Demand Rules and Horizon-Dependent Parameters

Forecast Weeks	Condition	Base	Direction
1–2	hype < 0.12, persistence < 0.18, momentum < -0.22	-1.00%	Bearish
	hype < 0.22, persistence < 0.19	-0.10%	Bearish
	hype < 0.22, persistence < 0.22	-0.20%	Bearish
	hype < 0.22, persistence < 0.24	-0.45%	Bearish
	momentum < -0.08, persistence > 0.22	$+1.40\%$	Bullish
	hype > 0.58, persistence > 0.48	$+1.10\%$	Bullish
3–4	persistence < 0.19	-0.15%	Bearish
	persistence < 0.27	-0.30%	Bearish
	momentum < -0.17	$+0.60\%$	Bullish
5–6	persistence < 0.20	-0.10%	Bearish
	persistence < 0.28	-0.25%	Bearish
	momentum < -0.14	$+0.45\%$	Bullish
7–8	persistence < 0.22	-0.08%	Bearish
	persistence < 0.30	-0.20%	Bearish
	momentum < -0.12	$+0.35\%$	Bullish
9–12	persistence < 0.32	-0.15%	Bearish
	momentum < -0.10	$+0.20\%$	Bullish

Base nudges are scaled by confidence and $\delta(h)$ before application. For weeks 1–2, four bearish conditions are defined with increasing severity: mild low-hype environments trigger corrections of -0.10% to -0.20%; a moderate persistence deficit produces -0.45%; and a combined extreme low-hype, low-persistence, negative-momentum state triggers the strongest downward correction of -1.00%. Two bullish rules also apply at this band: the Momentum Reversal rule $(+1.40\%)$ activates when momentum turns sharply negative against a high-persistence backdrop; the Strong Bull rule $(+1.10\%)$ fires under sustained high-hype, high-persistence conditions. For weeks 3–12, the rule structure simplifies: only persistence thresholds govern bearish nudges and a single momentum

TABLE II: Baseline and SMI-Nudged MASE and WQL Values of All Models Across Both Datasets[3]

	Walmart Dataset				M5 Dataset			
	Baseline		SMI-Nudged		Baseline		SMI-Nudged	
Models	**MASE**	**WQL**	**MASE**	**WQL**	**MASE**	**WQL**	**MASE**	**WQL**
ARIMA	2.436	0.1123	—	—	0.813	0.6253	—	—
STL-ARIMA	0.6386	0.0294	—	—	**0.6593**	**0.5071**	**0.6585**	**0.5065**
Prophet	**0.5424**	**0.025**	**0.5268**	**0.0243**	0.6714	0.5164	—	—
LSTM	0.7826	0.0641	—	—	0.7853	0.604	—	—
Chronos	**0.4759**	**0.0219**	**0.431**	**0.0199**	0.6612	0.5086	0.660	0.5076

threshold governs bullish nudges, with both thresholds and base magnitudes relaxing progressively across bands. A band-level floor cap prevents bearish nudges from exceeding a safe downward bound in each horizon range. No floor cap is applied for weeks 9–12 [5].

$$base = \max(base,\ \ell), \quad \ell = \begin{cases} 0.008 & \text{if } confidence < 0.72 \\ 0.0035 & \text{otherwise} \end{cases} \tag{5}$$

where ℓ is the minimum upward lift applied when persistence < 0.25, regardless of the assigned regime.

IV. Results and Analysis

The SMI–Nudger was evaluated using two complementary metrics. MASE measures point forecast accuracy by scaling mean absolute error against a seasonal naïve baseline; values below 1.0 indicate the model outperforms naïve.

$$MASE = \frac{1}{H} \sum_{t=1}^{H} \frac{|\hat{y}_t - y_t|}{\frac{1}{n-m} \sum_{t=m+1}^{n} |y_t - y_{t-m}|} \tag{6}$$

where n is the total number of observations in the training period

m is the seasonal period

H is the forecast horizon (number of weeks ahead)

y_t is the actual observed value at time t, and

$\hat{y}_t$ is the predicted value (point forecast) at time t

WQL evaluates probabilistic calibration across quantile levels $q \in \{0.1, 0.5, 0.9\}$ using pinball loss, penalising both overconfident and underconfident interval forecasts.

$$WQL_q = \frac{2}{n} \sum_{t=1}^{n} \max\left\{q(y_t - \hat{y}_{t,q}),\ (q-1)(\hat{y}_{t,q} - y_t)\right\} \tag{7}$$

where n is the total number of observations in the training period

y_t is the actual observed value at time t, and

$\hat{y}_t$ is the predicted value (point forecast) at time t

$\hat{y}_{t,q}$ is the quantile-specific predicted value at time t, and

q is the quantile level

A. MASE and WQL Analysis

Table II reports baseline and SMI-nudged MASE and WQL values across both datasets.

[3]Only the best performing model out of ARIMA, STL-ARIMA, Prophet, and LSTM was nudged, for each dataset.

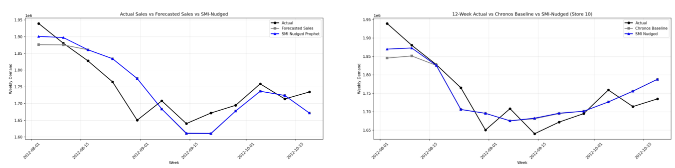
On the Walmart dataset II, Chronos achieves the strongest baseline (MASE = 0.4759, WQL = 0.0219). SMI-nudging reduces Chronos MASE by 9.44% (0.431) and WQL by 9.13% (0.0199); Prophet improves by 3.37% on MASE (0.5268) and 2.80% on WQL (0.0243).

On the M5 dataset II, STL-ARIMA leads baselines (MASE = 0.6593, WQL = 0.5071), narrowly ahead of Chronos (0.6612, 0.5086). Nudging yields marginal gains: STL-ARIMA MASE reduces by 0.12% (0.6585) and Chronos by 0.18% (0.660); WQL follows the same pattern. The bounded Nudger design produces no degradation on any model, confirming robustness under noisy or ambiguous momentum signals.

B. Actual vs. Forecasted vs. Nudged Sales

Walmart Dataset: Prophet Model 4a Nudging corrects Prophet's over-smoothing at turning points, improving tracking of sharp drops and partial recoveries in early and mid-horizon windows.

Walmart Dataset: Chronos Model 4b The Chronos baseline requires only moderate correction; nudged outputs maintain stable, coherent curves across mid-horizon dips and late-horizon rebounds.



(a) Prophet Model (b) Chronos Model

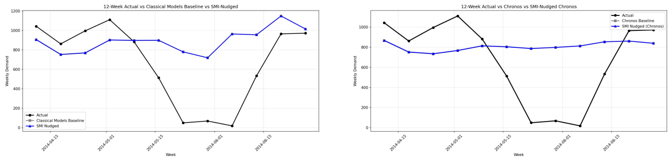
Fig. 4: Actual vs. Forecasted vs. Nudged Sales — Walmart Dataset

M5 Dataset: STL-ARIMA Model 5a STL-ARIMA underestimates extreme drop magnitudes; nudging partially compensates around high-shock weeks, though improvement is corrective rather than structural.

M5 Dataset: Chronos Model 5b Nudging introduces controlled adjustments that improve local fit across the early decline, mid-horizon low, and late rebound segments without amplifying noise.

C. Computational Efficiency

Table III summarises memory usage, compilation time, and FLOPs across all models. The SMI–Nudger adds no



(a) STL-ARIMA Model

(b) Chronos Model

Fig. 5: Actual vs. Forecasted vs. Nudged Sales — M5 Dataset

computational overhead beyond the baseline, operating as pure arithmetic over 8-week windows [5].

TABLE III: Computational Efficiency Across Models

Model	Memory (GB)	Compile Time (s)	FLOPs (MFLOPs)
ARIMA	0.003	0.528	0.13
STL-ARIMA	0.002	0.520	0.26
Prophet	0.002	0.125	6.50
LSTM	0.072	1.305	123,615.00
Chronos	0.007	1.709	62.4×10^6

V. Conclusion

The SMI–Nudger framework is an interpretable, rule-based post-processing layer that consistently improves retail demand forecasts without retraining. On the Walmart dataset, Chronos with SMI achieves a 9.44% relative reduction in MASE and 9.13% in WQL. On sparse M5 series, improvements are marginal but consistent, and no model's performance is degraded. The same nudging logic produces directional improvements across STL-ARIMA, Prophet, and Chronos without model-specific modifications, validating the model-agnostic design.

Three directions extend this framework. First, hierarchical momentum computation at item, store, and category levels would capture local demand shifts invisible at the aggregate. Second, replacing hand-crafted Nudger thresholds and decay factors via lightweight optimisation (grid search or Bayesian optimisation) would improve generalisation to locally shifting demand patterns. Third, incorporating promotional calendars, weather, or macroeconomic indicators into the SMI confidence score would improve correction during atypical demand periods. Translating MASE and WQL improvements into operational metrics — stockout rate reduction, overstock holding cost savings, and order quantity optimisation — via replenishment policy simulation remains an open evaluation direction [5].

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